

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and trained **deep learning networks** for scene understanding in automotive environments, applying **3D object detection** and **pedestrian classification** techniques.
- Selected and refined appropriate **datasets** and data representation methods for training computer vision models, focusing on **image classification** and **trajectory mapping**.
- Implemented **Python** scripts to automate data acquisition, ensure data integrity, and build robust **data pipelines** for machine learning model training.
- Performed **statistical analysis** and fine-tuning of model performance using test results, achieving a **15% improvement** in classification accuracy for vehicle detection.
- Contributed to the development of **state-of-the-art DL algorithms** for automated driving systems, integrating sensor fusion outputs into a detailed environmental map.
- Collaborated with university counterparts on research initiatives, staying abreast of technological developments in **deep learning** and its application in the automotive sector.

SKILLS

Deep Learning Frameworks – Keras, PyTorch, scikit-learn

Computer Vision – OpenCV, 3D object detection, Pedestrian classification, Trajectory mapping, Lane detection

Programming Languages – Python, English

Data Handling – Data acquisition, Data integrity, Data pipelines, Statistical analysis, Data modeling

Software Development – Linux, Git, Version control, Software architecture, Embedded platforms

PROJECTS

Automotive Scene Understanding Model (HCL Technologies)

- Developed a **deep learning model** using **PyTorch** for real-time classification of road objects and surfaces from vehicle sensor data.
- Engineered feature extraction pipelines in **Python** and **OpenCV**, processing ~1.5 million images to create a comprehensive training dataset.
- Trained the classification model on annotated image data, achieving an **88% accuracy** in identifying vehicles and pedestrians in diverse driving conditions.

Embedded Vision System Integration (HCL Technologies)

- Integrated computer vision algorithms onto an **embedded Linux platform**, optimizing for resource-constrained automotive hardware.
- Developed **Python** scripts for data preprocessing and model deployment workflows, ensuring compatibility with the target system architecture.
- Collaborated with a small team to refine the perception system, improving **collision mitigation** predictions by 18% through enhanced object detection.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)